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Assignment 1

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Problem 1

Water flows steadily through a fire hose and nozzle. The hose is 35 mm in diameter and the nozzle tip is 25 mm in diameter; water gauge pressure in the hose is 510 kPa, and the stream leaving the nozzle is uniform. The exit speed and pressure are 32 m/s and atmospheric, respectively.

Find the force transmitted by the coupling between the nozzle and hose. Indicate whether the coupling is in tension or compression. State the assumptions made and sketch the situation.

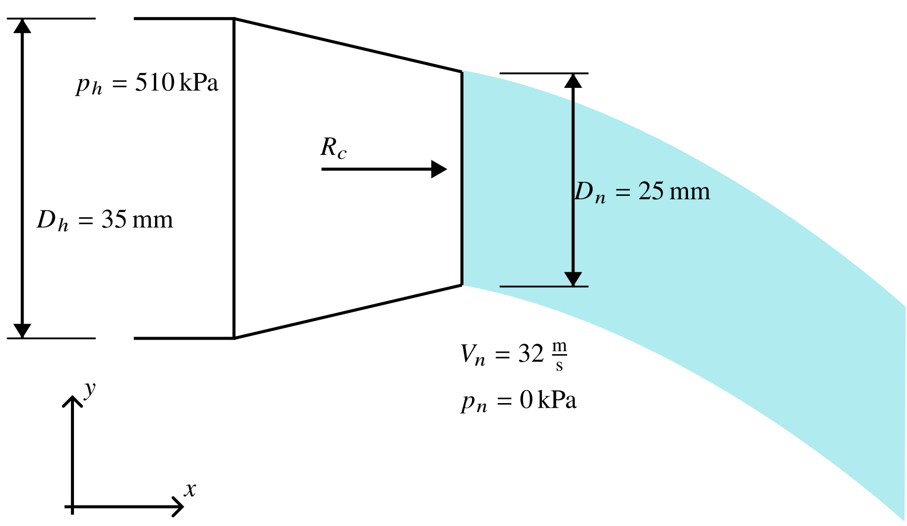


Figure 1.1: Sketch of the coupling.

Assumptions

- Steady flow
- Uniform flow at interfaces between coupling, hose, and nozzle
- Incompressible and frictionless liquid with negligible mass

Derivation

The momentum equation for an inertial control volume (as is the case for the coupling) is

$$\boldsymbol{F} = \boldsymbol{F}_B + \boldsymbol{F}_S + \frac{\partial}{\partial t}\int_{\text{CV}} \boldsymbol{V}\rho\,\mathrm{d}V + \int_{\text{CS}} \boldsymbol{V}\rho\boldsymbol{V}\cdot\mathrm{d}\boldsymbol{A}.$$

In this case the flow is steady, hence $\frac{\partial}{\partial t} = 0$, hence the $\boldsymbol{F}_B = 0$. Reducing the vector equation to a scalar equation in the x-direction (defined on Figure 1.1) we obtain

$$F_x = F_{S_x} = \int_{CS} u\rho\boldsymbol{V}\cdot\mathrm{d}\boldsymbol{A}.$$

As the flow is assumed uniform at both the inlet and outlet the integral further reduces to a sum as

$$F_x = F_{S_x} = \sum_{\text{CS}} u\rho\boldsymbol{V}\cdot\boldsymbol{A} \tag{1.1}$$

The forces in the x-direction on the control volume are the pressure at the interface between the hose and the coupling ($+p_h A_h$), the pressure at the nozzle ($-p_n A_n$), and the reaction on the coupling R_x. From Equation (1.1) we get

$$R_x + p_h A_h - p_n A_n = \rho Q \left(V_n - V_h\right) \tag{1.2}$$

As we are using gauge pressures $p_n = 0\,\text{kPa} \implies p_n A_n = 0\,\text{N}$, therefore Equation (1.2) reduces to

$$R_x + p_h A_h = \rho Q \left(V_n - V_h\right) \tag{1.3}$$

The continuity equation prescribes that the flow rate $Q = VA = \text{const}$. Hence,

$$Q = A_n V_n \tag{1.4a}$$

$$V_h = \frac{Q}{A_h} = \frac{A_n V_n}{A_h} \tag{1.4b}$$

Substituting Equation (1.4) into Equation (1.3) and solving for R_x we obtain

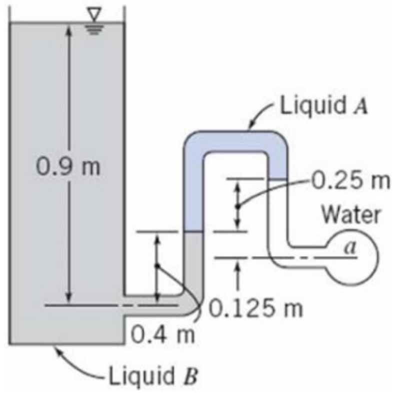
$$\begin{aligned}
R_x + p_h A_h &= \rho A_n V_n \left(V_n - \frac{A_n V_n}{A_h}\right) \\
R_x &= \rho A_n V_n^2 \left(1 - \frac{A_n}{A_h}\right) - p_h A_h \\
R_x &= 1000\,\frac{\text{kg}}{\text{m}^3} \cdot \left(\pi \cdot (12{,}5\,\text{mm})^2\right) \cdot \left(32\,\frac{\text{m}}{\text{s}}\right)^2 \left(1 - \frac{(12{,}5\,\text{mm})^2}{(17{,}5\,\text{mm})^2}\right) - 510\,\text{kPa} \cdot \pi \cdot (17{,}5\,\text{mm})^2 \\
R_x &= -244{,}48\,\text{N}.
\end{aligned}$$

The negative sign (with $+x$ being to the right as shown on Figure 1.1) means the coupling reaction force points to the left, and the joint will therefore be in a tension of about 245 N.

$$\boxed{R_c = 245\,\text{N} \quad (\text{Tension})}$$ Result

Problem 2

Determine the gauge pressure in kPa at point a, if liquid A has SG = 1,20 and liquid B has SG = 0,75. The liquid surrounding point a is water, and the tank on the left is open to the atmosphere.



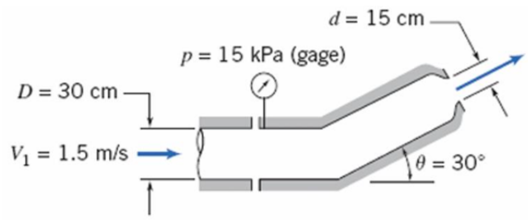
Problem 3

Determine the stream function ψ that will yield the velocity field.

$$\boldsymbol{V} = 2y\,(2x+1)\,\hat{\imath} + \left(x\,(x+1) - 2y^2\right)\hat{\jmath}\,V.$$

Problem 4

Water flows steadily through the nozzle, shown in the figure below, discharging to atmosphere. Calculate the horizontal component of force in the flanged joint. Indicate whether the joint is in tension or compression. Use density of water = 1000 $\frac{\text{kg}}{\text{m}^3}$.



Problem 5

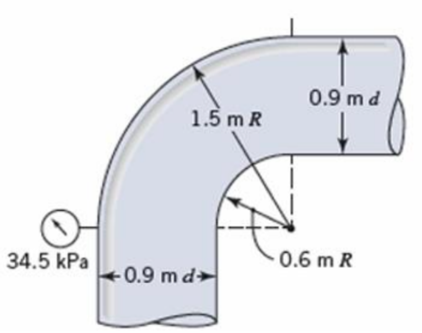
The velocity components for an incompressible steady flow field are given below

$$u = A\left(x^2 + z^2\right) \qquad v = B\,(xy + yz)\,.$$

Determine the z-component of velocity for steady and unsteady flow.

Problem 6

The water flow rate through the vertical bend shown in the figure below is 2,83 $\frac{\text{m}^3}{\text{s}}$. Calculate the magnitude, direction, and location of the resultant force of the water pipe bend. Use density of water = 999 $\frac{\text{kg}}{\text{m}^3}$.



Problem 7

A viscous liquid is sheared between two parallel disks of radius R, one of which rotates while the other is fixed. The velocity field is purely tangential, and the velocity varies linearly with z from $V_\phi = 0$ at $z = 0$ (the fixed disk) to the velocity of the rotating disk at its surface ($z = h$).

Derive an expression for the velocity field between the disks.